

GHSS Regional Youth Innovation Challenge 2026

YCD Farmer Guide

AI-Powered Agricultural Support Platform

APP PROTOTYPE

Interactive User Flow & Screen Showcase

20+

Screens

6

User Flows

AI

Powered

2

Languages

Agritech Domain | Cameroon | March 2026

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App Prototype — Table of Contents

This prototype document showcases the **YCD Farmer Guide** mobile application through annotated user flows. Each section presents a key user journey with actual app screens, navigation paths, and feature highlights.

1	Home & Dashboard	Main hub with quick-access feature cards
2	AI Disease Detection	Camera → AI Analysis → Treatment Plan
3	AI Farming Assistant	Chat-based AI for farming advice
4	Weather & Climate	Real-time localized weather forecasts
5	Agricultural Marketplace	Browse → Product Details → Cart → Payment
6	Expert Consultation	Find experts → Expert profile → Live chat
7	Profile & Settings	User management, language, preferences
8	Admin Dashboard	Platform management, analytics, user control

Technology Stack

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Frontend:

React Native + Expo SDK 55

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Backend:

Node.js + Express.js (34 Controllers, 43 Models)

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AI Engine:

Groq LLaMA 4 Scout + Whisper V3 (Voice)

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Database:

PostgreSQL + PostGIS (Geospatial)

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Cloud:

Railway + Cloudinary + Upstash Redis

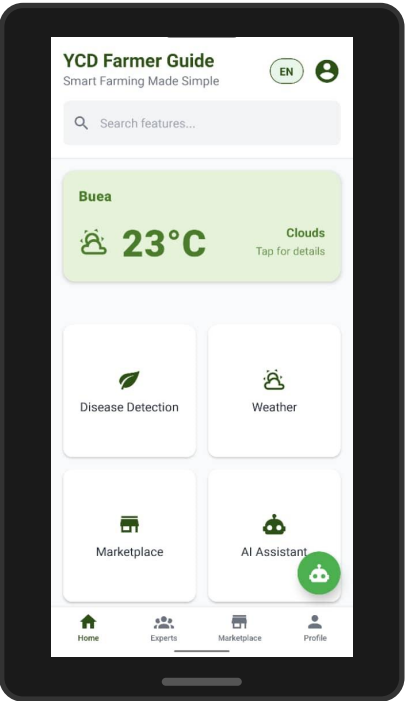
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Security:

JWT Auth, Rate Limiting, Input Sanitization

■ Flow 1 — Home Dashboard

The home screen serves as the central hub, providing quick access to all core features through intuitive cards. It displays weather summaries, recent marketplace listings, and personalized AI recommendations.

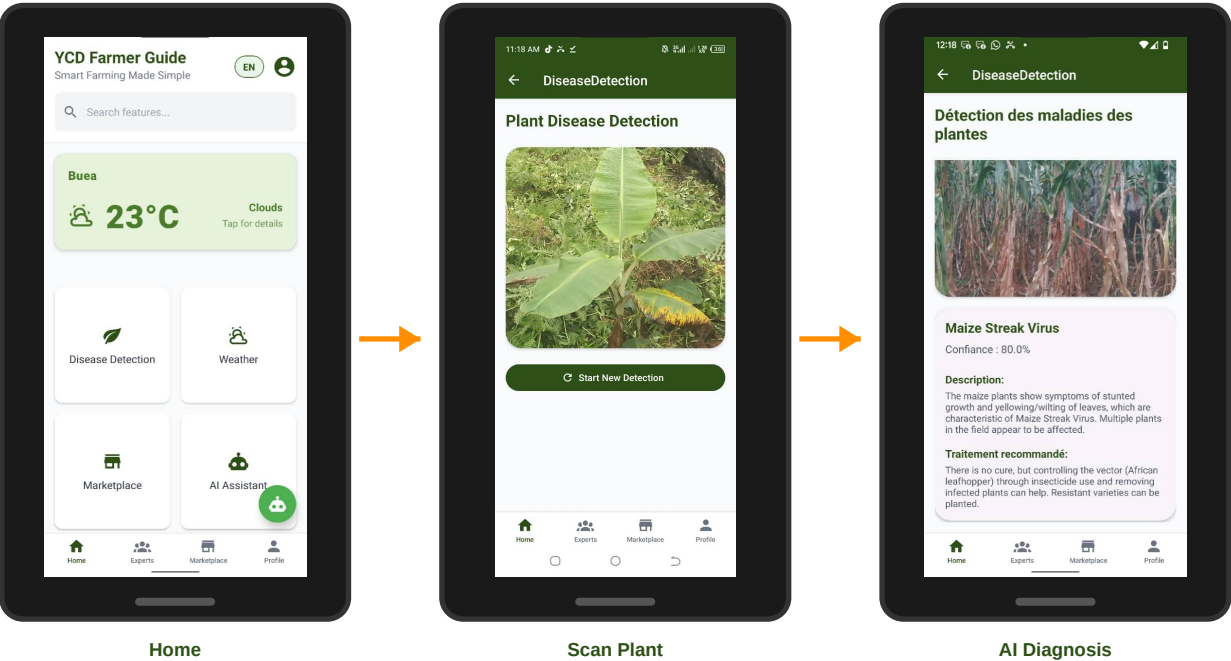


Home Screen

- **Quick Actions:** One-tap access to Disease Detection, Weather, AI Chat, Marketplace
- **Dashboard Stats:** Crop health overview, market trends, weather alerts
- **Notifications:** Real-time alerts for expert replies, forum posts, orders
- **Location-Aware:** Content personalized to farmer's region in Cameroon

■ Flow 2 — AI Disease Detection

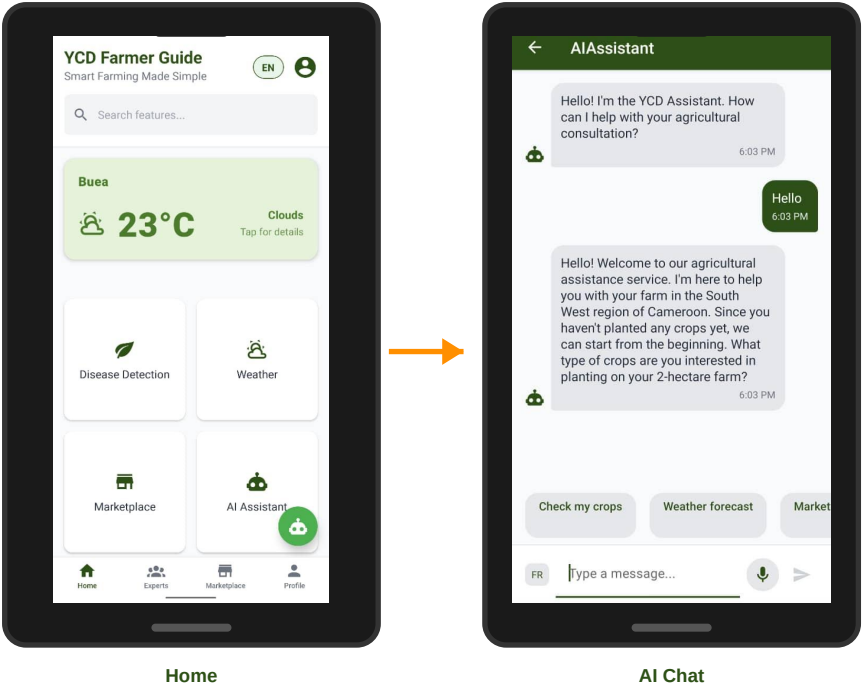
Farmers photograph sick plants using their smartphone camera. The AI model identifies diseases with high accuracy and provides immediate treatment recommendations including organic and chemical options with local availability.



- **Camera Scan:** Take or upload a photo of the affected plant
- **AI Analysis:** Groq LLaMA 4 Scout identifies disease with confidence score
- **Treatment Plan:** Organic & chemical remedies with local product recommendations
- **History:** Track past scans to monitor crop health over time

■ Flow 3 — AI Farming Assistant

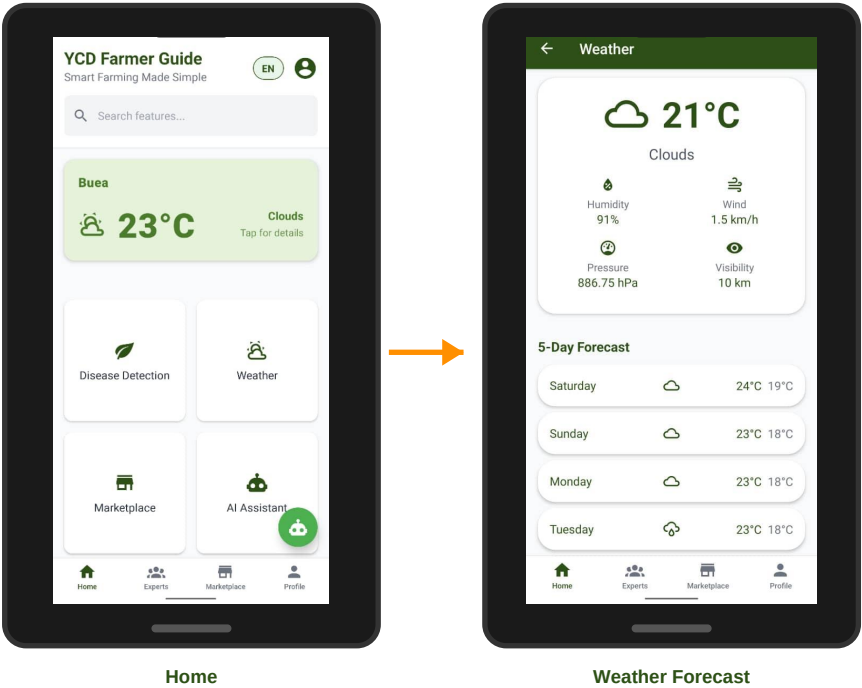
An intelligent chat assistant that answers farming questions in natural language. Supports voice input (Whisper V3) and provides region-specific advice on planting schedules, soil management, pest control, and more.



- **Natural Chat:** Ask questions in English or French, get expert-level answers
- **Voice Input:** Whisper V3 speech-to-text for hands-free use in the field
- **Context-Aware:** Advice tailored to Cameroon's crops, climate, and soil types
- **Farming Guidance:** Step-by-step crop cultivation guides and best practices

■ ■ Flow 4 — Weather & Climate Intelligence

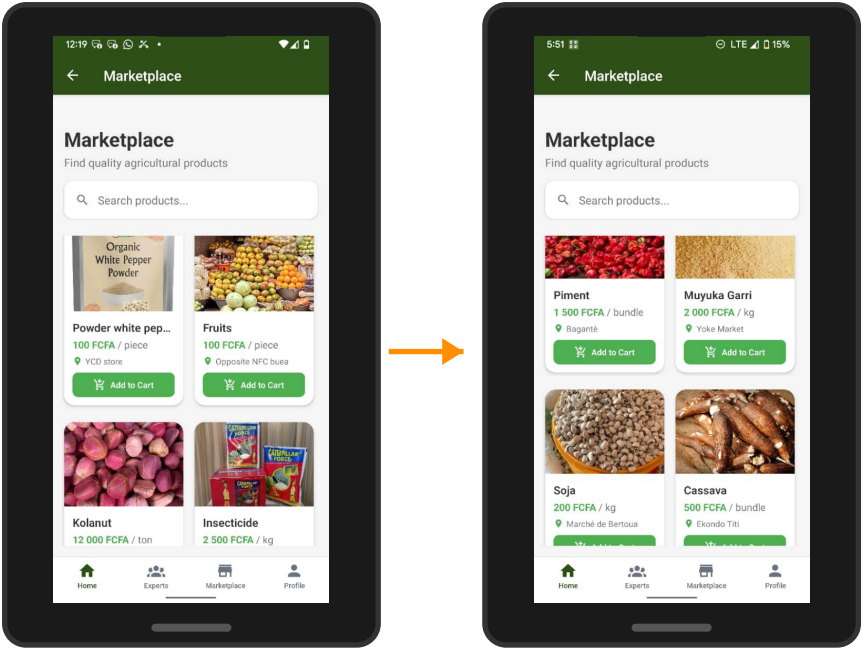
Real-time localized weather forecasts help farmers plan planting, irrigation, and harvesting. The system uses OpenWeather API with PostGIS for precise location-based forecasts across Cameroon's diverse climate zones.



- ■ **Current Conditions:** Temperature, humidity, wind speed, precipitation
- **7-Day Forecast:** Plan farming activities with weekly weather outlook
- ■ **Weather Alerts:** Extreme weather warnings for crop protection
- ■ **Region-Specific:** Forecasts for all 10 regions of Cameroon

■ Flow 5 — Agricultural Marketplace

A complete e-commerce platform connecting farmers directly with buyers. Farmers can sell produce, seeds, and tools; buyers browse with category filters. Supports Mobile Money (MTN MoMo, Orange Money) for payments.



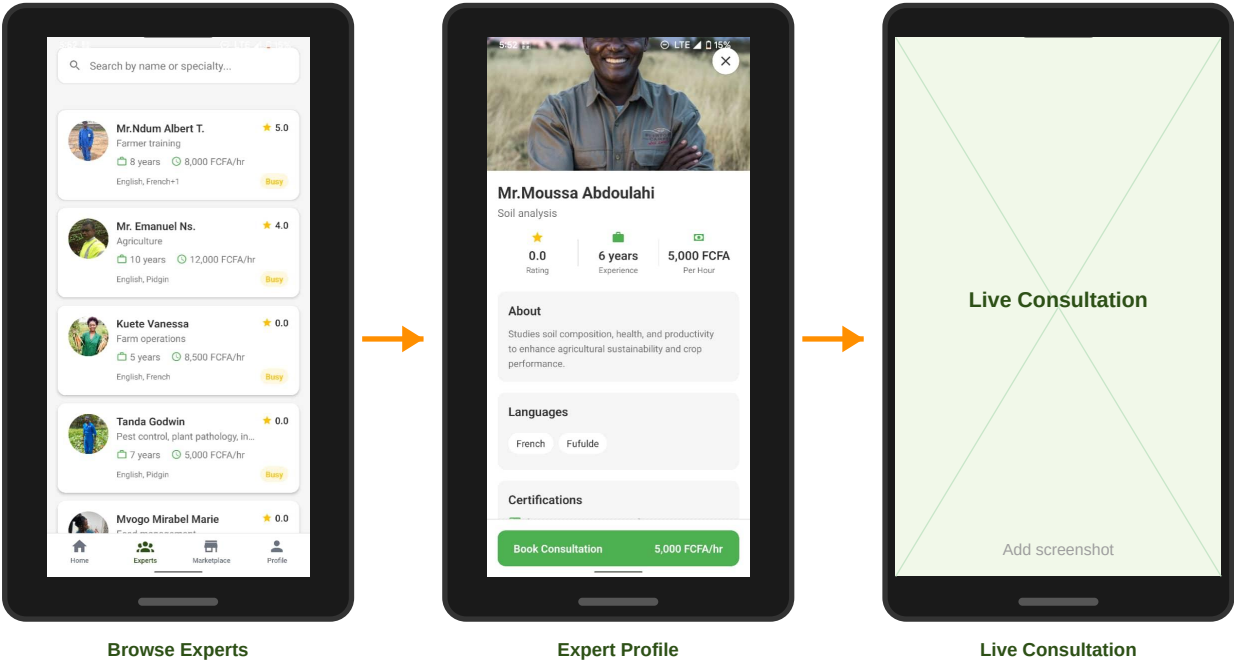
Marketplace

Product Details

- **Product Listings:** Seeds, fertilizers, tools, and fresh produce
- **Fair Pricing:** Eliminates middlemen — farmers earn up to 50% more
- **Mobile Money:** MTN MoMo & Orange Money integration
- **Order Tracking:** Full order history and delivery status

■■■ Flow 6 — Expert Consultation

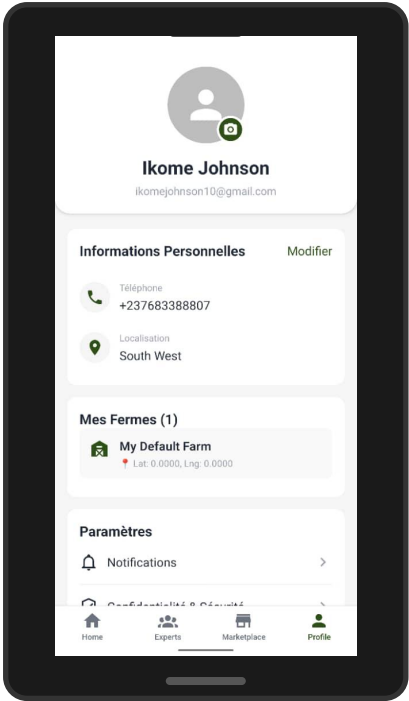
Farmers connect with verified agricultural experts for personalized advice. Experts include agronomists, soil specialists, and pest control professionals. Live chat with text, images, and real-time consultation tracking.



- **Expert Profiles:** Verified credentials, specialization, ratings, and availability
- **Live Chat:** Real-time text and image-based consultations
- **Socket.IO:** Instant notifications when experts respond
- **Rating System:** Farmers rate consultation quality for accountability

■■ Flow 7 — Profile & Settings

Users manage their profile, farm details, language preferences, and app settings. The profile stores farm location for region-specific content, and supports switching between English and French.

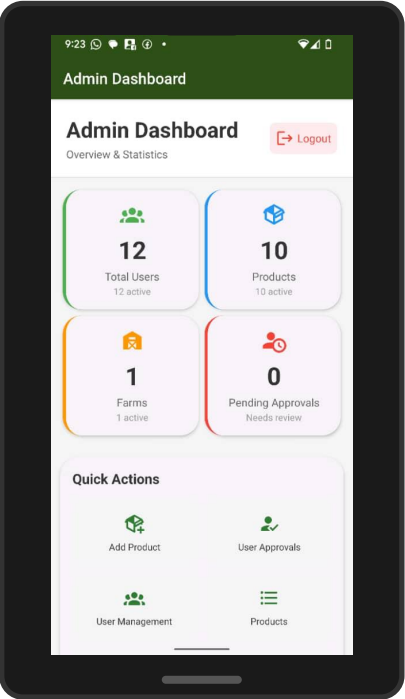


My Profile

- **Farm Profile:** Name, region, farm size, main crops
- **Language Switch:** Toggle between English and French instantly
- **Security:** Change password, manage sessions
- **Activity History:** View past scans, orders, consultations

■ Flow 8 — Admin Dashboard

Administrators have a dedicated dashboard for platform management. Monitor user activity, approve new registrations, manage products and experts, moderate forums, and view real-time analytics on platform usage.



Admin Dashboard

- **Analytics:** Real-time platform stats — users, scans, orders, revenue
- **User Management:** Approve registrations, manage roles and permissions
- **Product Control:** Add, edit, and moderate marketplace listings
- **Expert Oversight:** Verify and manage agricultural expert accounts
- **Forum Moderation:** Create categories, moderate discussions

■ Impact & Innovation Summary

Feature	Innovation	Impact on Farmers
AI Disease Detection	LLaMA 4 Scout vision model	Reduces crop loss by 20-40%
Voice AI Assistant	Whisper V3 speech-to-text	Accessible to low-literacy users
Agricultural Marketplace	Direct farmer-to-buyer platform	Eliminates middleman exploitation
Expert Consultation	Real-time Socket.IO chat	1:3000 ratio overcome digitally
Weather Intelligence	PostGIS + OpenWeather API	Climate-smart farming decisions
Community Forums	Region-based peer network	Scalable knowledge sharing

Target Beneficiaries: 5+ million smallholder farmers across Cameroon, starting with the South-West, Littoral, and West regions.

Scalability: Designed for multi-country expansion across sub-Saharan Africa. AI models can be fine-tuned for local crops and diseases in any region. Roadmap includes Pidgin, Ewondo, Fulfulde, Duala, and Bamiléké language support.